

Part 1 胡教材 P40-P42 2-4. $[v]=L T^{-1}$; $[g]=L T^{-2}$; $[h]=L$ $\therefore v \propto \sqrt{gh}$

2-5: $[T]=T$; $[m]=M$; $[k]=M \cdot T^{-2}$; $[a]=L$ $\therefore T \propto \sqrt{\frac{m}{k}}$

2-6 为简化运算, 我们采用经典物理模型, 记光子质量为 m_0 , 系用平面极坐标系, 以黑洞为原点

$\therefore -\frac{Gmm_0}{R^2} = -m\frac{v^2}{R}$ 当 $v=c$ 时, R 为视界半径 $\therefore Gm = c^2 R$, 即为所求.

2-11: 我们采用平面直角坐标系. 下面对 m 与 M 进行受力分析: (我们不再考虑 m 与 M 发生静摩擦的情况)

M : m :
对 m : $\begin{cases} N_1 = mg \\ f_1 = \mu N_1 = \mu mg \end{cases}$ 对 M : $\begin{cases} N_1' = N_1 = mg \\ N_2' = N_1' + N_2 = (m+M)g \\ f_1' = f_1 = \mu mg \\ f_2' = \mu N_2 = \mu(m+M)g \end{cases}$
解得: $a_m = mg$ $a_M = \frac{F}{M} - \frac{\mu(m+M)g}{M}$ $a_{相对} = \frac{1}{M}(F - \mu(2m+2M)g)$

当 m 落至桌面后, $a_m' = -mg$ $\therefore$ 加速度最多在 $L-1$, 即: $\frac{1}{2}at^2 \leq L-1$

$\therefore$ 我们有 $\begin{cases} a_m < a_M \text{ (滑动条件)} \\ a_{相对} \leq L-1 \end{cases}$ 解得: $F \geq 2mg(m + \frac{ML}{L-1})$
 $\frac{1}{2}at^2 \leq L-1$ (约束条件)

2-12, 以向下为正, 建立一维坐标系, 我们考虑从释放小环至加速结束的过程.

进行受力分析: m_1 : m_2 :
 $\therefore \begin{cases} a_1 = g + \frac{f}{m_1} \\ a_2 = g - \frac{f}{m_2} \end{cases}$
 $\therefore \begin{cases} v_1 = at \\ v_2 = v + at \end{cases}$ $\frac{1}{2}v_1 = v_2 \therefore v + at_0 = a_1 t_0 \Rightarrow t_0 = \frac{m_1 m_2}{m_1 + m_2} \frac{v}{f}$ $v_{相对} = a_1 t_0 = (g + \frac{f}{m_1}) (\frac{m_1 m_2}{m_1 + m_2} \frac{v}{f})$
 $v_{相对} = v_2 - v_1 = v + (a_2 - a_1)t \therefore x = \int_0^{t_0} v_{相对} dt = vt_0 + (a_2 - a_1) \frac{t_0^2}{2}$
 $= \frac{m_1 m_2}{m_1 + m_2} \frac{v^2}{f} + (-\frac{f}{m_2} - \frac{f}{m_1}) \cdot \frac{1}{2} (\frac{m_1 m_2}{m_1 + m_2})^2 \frac{v^2}{f^2} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \frac{v^2}{f}$

2-13. 情况1: 记绳上张力为 T , 体系加速度为 a_0 , 以向右向上为正建系.

受力分析: 对 m_1 : 对 m_2 : 对 M :
 $\therefore \begin{cases} N_1 = m_1 g \\ T_0 = m_1 a_0 \end{cases} \begin{cases} T_0 = m_1 g \\ N_2 = m_2 a_0 \end{cases} \begin{cases} N = T_0 + m_1 g + M g \\ F - T_0 - m_2 a_0 = M a_0 \end{cases} \Rightarrow a_0 = \frac{m_2}{m_1} g$
 $\therefore \begin{cases} N = (m_1 + m_2 + M)g \\ F = (m_1 + m_2 + M) \frac{m_2}{m_1} g \end{cases}$

情况2: 记绳上张力为 T_1 , m_1 与 m_2 加速度大小为 a_1 , 建系同上.

受力分析: 对 m_1 : 对 m_2 : 对 M :
 $\therefore \begin{cases} N_1 = m_1 g \\ T_1 = m_1 a_1 \end{cases}, m_2 g - T_1 = m_2 a_1, \begin{cases} N = m_1 g + T_1 + M g \\ F = T_1 \end{cases}$
解得: $a_1 = \frac{m_2}{m_1 + m_2} g$ $F = \frac{m_1 m_2}{m_1 + m_2} g$ $N = (m_1 + M + \frac{m_1 m_2}{m_1 + m_2}) g$

2-17 向下为正, 建系

由受力分析 $\therefore (mg - f) = ma$ $\therefore mg - cv = m \frac{dv}{dt}$ $\therefore \frac{1}{m} dt = \frac{dv}{mg - cv}$

两侧不定积分: $C + \frac{1}{m} t = -\frac{1}{c} \ln(mg - cv)$ $t=0$ 时, $v=v_0 \therefore C = -\frac{1}{c} \ln(mg - cv_0)$

$\therefore \frac{1}{m} t = -\frac{1}{c} \ln \frac{mg - cv}{mg - cv_0} \therefore \frac{mg - cv}{mg - cv_0} = \exp(-\frac{c}{m} t)$

2-18. (1) 由受力分析: 对 Monkey: 对箱子: $\therefore$ 当 $F \geq mg$, 即 $N=0$ 时箱子会落地.

$\therefore F - mg = ma \geq mg - mg \therefore a \geq \frac{1}{3}g$ g 为重力加速度.

(2) 由受力分析: 对 Monkey: 对箱子:
 $\begin{cases} T - mg = ma \\ mg - T = Ma \end{cases} \therefore \frac{1}{m} T - g = g - \frac{1}{M} T \therefore T = \frac{mM}{m+M} g$

$a = \frac{2M}{m+M} g - g = \frac{M-m}{m+M} g$ 即为所求.

Part 2: 吴教材 P79 2-11 以关闭发动机时船前进的方向为正

(1). 由牛顿第二定律: $ma = -kv^2$ $\therefore \frac{dv}{dt} = -\frac{k}{m} v^2 \therefore -\frac{dv}{v^2} = \frac{k}{m} dt$

两侧不定积分: $\frac{1}{v} = \frac{k}{m} t + C$ $t=0$ 时, $v=v_0 \therefore C = \frac{1}{v_0} \therefore v = \frac{1}{\frac{k}{m} t + \frac{1}{v_0}} = \frac{m v_0}{k v_0 t + m}$

(2). 由上问: $\frac{dx}{dt} = \frac{m v_0}{k v_0 t + m} \therefore dx = \frac{m v_0}{k v_0 t + m} dt \therefore x = \frac{m}{k} \ln(k v_0 t + m) + C$ $t=0$ 时, $x=0$

$\therefore C = -\frac{m}{k} \ln m \therefore x = \frac{m}{k} \ln(\frac{k v_0}{m} t + 1) = \frac{m}{k} \ln(\frac{k v_0}{m} t + 1)$

(3). $x = \frac{m}{k} \ln(\frac{k v_0 t + m}{m}) = \frac{m}{k} \ln(\frac{v_0}{v})$, 即为所求.

2-12. 先对小球做受力分析
 $\begin{cases} N = m a_n = m \frac{v^2}{R} \\ f = \mu N = \mu m \frac{v^2}{R} = -m a_t \end{cases} \therefore a_t = -\mu \frac{v^2}{R}$

我们对上面得到的切向加速度与速率的关系: $\frac{dv}{v^2} = -\frac{\mu}{R} dt$, 两侧不定积分.

$\therefore \frac{1}{v} = \frac{\mu}{R} t + C$ $t=0$ 时, $v=v_0 \therefore C = \frac{1}{v_0} \therefore v = \frac{1}{\frac{\mu}{R} t + \frac{1}{v_0}} = \frac{R v_0}{\mu v_0 t + R}$

$\therefore ds = \frac{R v_0}{\mu v_0 t + R}$, 同理可求得 $s = \frac{R}{\mu} \ln(1 + \frac{\mu v_0 t}{R})$

2-14. 为了解题, 我们采用与管共转的旋转坐标系: (自然系) (向外, 向转动方向为正)

$\omega \omega$ $F_c = m \omega^2 r$ (惯性离心力)

(1) $ma = m \omega^2 r \therefore \frac{d^2 r}{dt^2} = \omega^2 r \therefore \frac{d^2 r}{dt^2} \cdot \frac{dr}{dt} = r \frac{d^2 r}{dr} = \omega^2 r \therefore \frac{1}{2} v_r^2 = \frac{1}{2} \omega^2 r^2 + C$

$\therefore C=0 \therefore v_r^2 = \omega^2 r^2 \therefore v_r = -\omega r$

(2) 由上问: $\frac{dr}{dt} = -\omega r \therefore \ln|r| = -\omega t \therefore -\ln r = -\omega t \therefore t = \frac{\ln r}{-\omega}$

Part 3. 补充练习1: (1). 对 m_1 : 对 m_2 :
 $\therefore \begin{cases} a_1 = (T + m_1 g) \sin \alpha / m_1 \\ a_2 = (m_2 g - T) / m_2 \end{cases}$ 考虑竖直分量 $\begin{cases} a_1' = (\frac{T}{m_1} + g) \sin \alpha \\ a_2' = (g - \frac{T}{m_2}) \end{cases}$

$a_1' = a_2' \Rightarrow (\frac{T}{m_1} + g) \sin \alpha = g - \frac{T}{m_2} \therefore T(\frac{\sin^2 \alpha}{m_1} + \frac{1}{m_2}) = g \cos^2 \alpha \therefore T = \frac{g \cos^2 \alpha}{\frac{\sin^2 \alpha}{m_1} + \frac{1}{m_2}} m_1 m_2$

(2) 将 m_1 与 m_2 视作整体, $\therefore a = g \sin \theta$ 在非惯性系(对岸), m_2 的受力平衡.

因此, T 必然指向与细杆垂直的方向
即细杆与竖直线夹角必须为 α

Part 4: 附加题. 我们采用当地的自然系, 使 x 轴指向正东, y 轴指向正北, z 轴向上

则 $\vec{\omega} = \omega \cos \phi \hat{j} + \omega \sin \phi \hat{k}$ $v_{zt} = \omega - g t$ $\vec{v}_2 = (0, 0, \omega - g t)$

$\therefore \vec{\omega} \times \vec{v} = (\omega \cos \phi)(\omega - g t) \hat{i} \therefore a_x = -2 \omega \cos \phi (\omega - g t)$

连续两次对 t 积分: $v_x(t) = -2 \omega \cos \phi (\omega t - \frac{1}{2} g t^2)$, $x(t) = -\omega \cos \phi (\omega t^2 - \frac{1}{3} g t^3)$ $\therefore x(1) = -\frac{8}{3} \sqrt{2} \cos \phi \omega \sqrt{\frac{h^3}{g}}$